# EECS 841: Computer Vision

## Lecture 3


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# Outline

- Conceptualizing images
- Fundamentals of Imaging: Chapter 2
    - Fundamental concepts of imaging.
    - A quick tour of natural vision systems, with particular attention paid to the human visual system.
    - Topics of image formation and acquisition, such as the pinhole camera model, lenses, sampling, and quantization.
    - A survey of alternative imaging modalities.
    - A detailed look at the electromagnetic spectrum.

# Conceptualizing Images

- A digital image is stored in the computer as a discrete array of values, which can be visualized either by:
  - considering the raw pixel values themselves arranged in a 2D lattice
  - equivalently as a height map, or 3D surface plot

- $I(x, y)$: evaluate the function at the position $(x, y)$.

# Conceptualizing Images

$$I = \begin{bmatrix} 3 & 8 & 0 \\ 2 & 9 & 4 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

- Set of pixels
  - The grayscale image: Example:

$$I = \begin{bmatrix} 3 & 8 & 0 \\ 2 & 9 & 4 \end{bmatrix}$$

  - The binary image: Example:

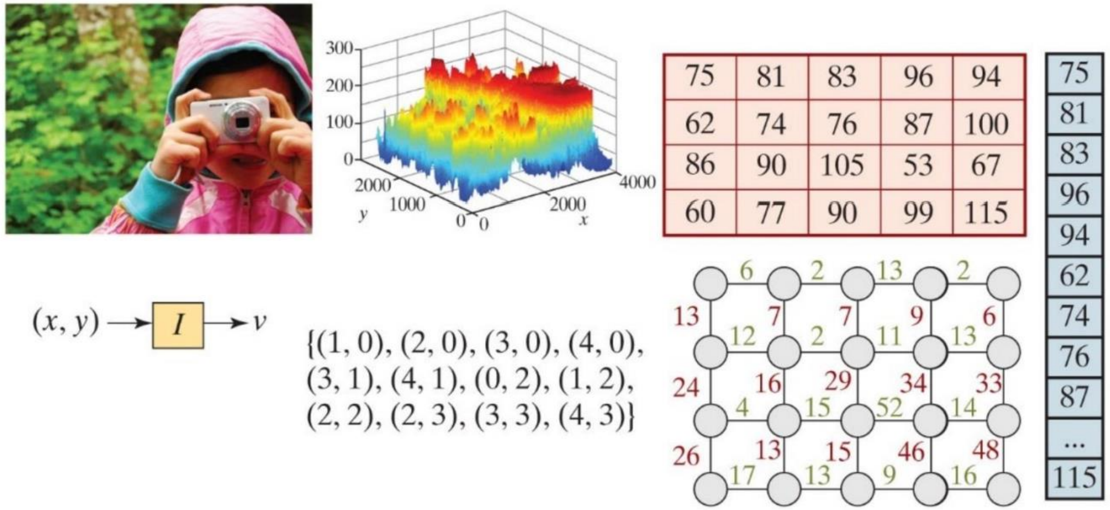
$$I = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

  - The following matrix **A** is an $m \times n$ matrix whose $(i, j)^{th}$ entry is given by $a_{ij}$

$$\mathbf{A}_{\{m\times n\}} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

# More ways: Conceptualizing Images

**Figure 1.6:** Different ways to visualize an image: as a picture, as a height map, as an array of values, as a function, as a set, as a graph, and as a vector. The 5 × 4 array is a small portion of the image; the set contains the coordinates of all pixels in the array whose value is greater than 80; and the weights of the edges in the graph are the absolute differences between values in the array.




- Image, Height map, 2D matrix, 1D Vector [Top row: Left to right]
- Function, Set, Graph [Bottom row: Left to right]

Guess the type of conceptualization!

- Image
- Height map: height represents intensity
- 2D matrix: matrix of intensity values
- 1D Vector: 1D vector of intensity values → can be row major or column major
- Function: image as a function defined on *x, y* grid → takes two input parameter to output intensity → *x,y* is domain, *I* is the image is function
- Set: This is a set of two elements which represents the location of the image such that the intensity there is >75 → [ex. (1,0)'s intensity is 81 → therefore it is in the set → (1,0) means column 1 row 0
- Graph: image as a graph → Each circle represents a pixel → each edge represents the difference on intensity value from the pixel next to it → Why graph: many cv uses graph

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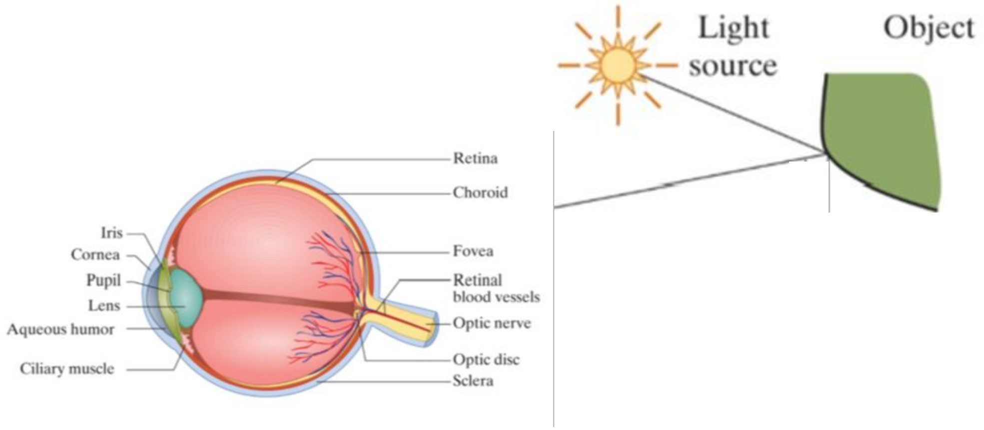
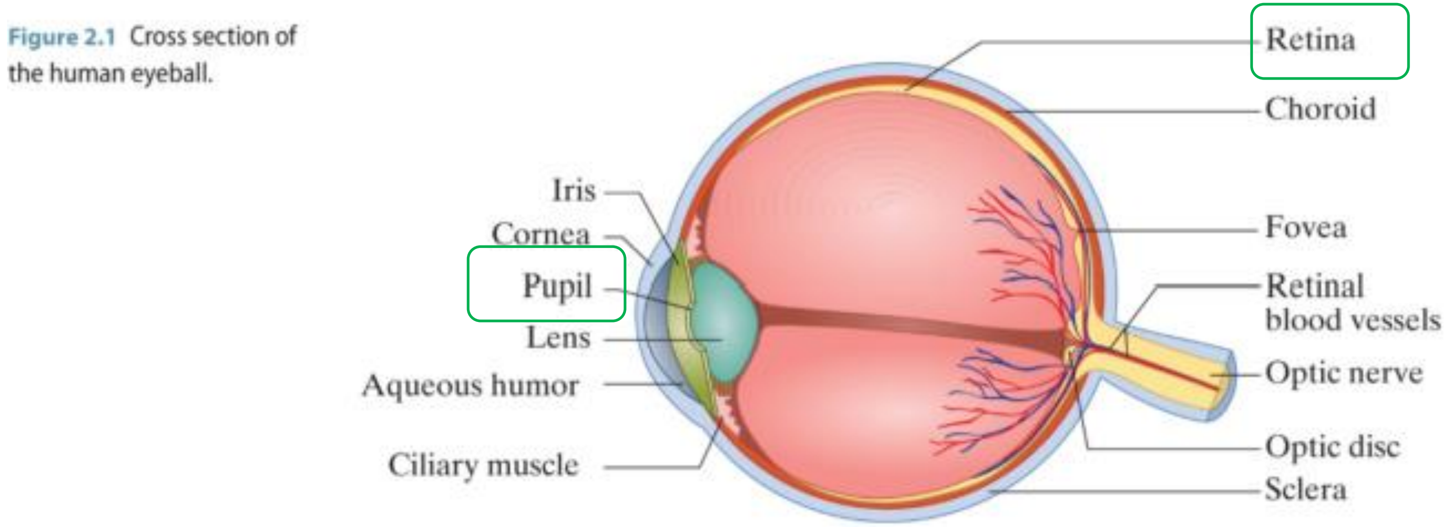
# Basis of Photo Creation In Human

How we see an image?



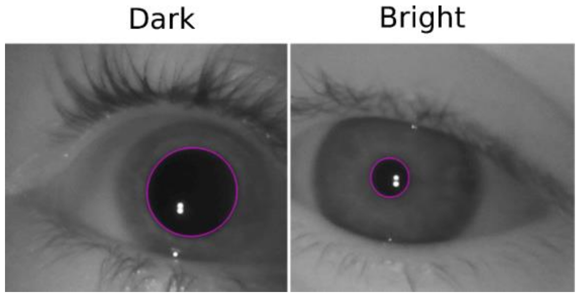

- **Photoreceptor:** The most fundamental component of any natural vision system.
  - When a single photon of light hits a single photoreceptor, it sets off a wonderfully complex chain of events that leads to the surprising ability to see.
  - The absorption of a photon causes a change in shape of a small organic molecule called *retinal*. This change in shape causes the larger protein, called *rhodopsin*, holding the retinal molecule to change shape.

# Human Eye




Human eye in brief:

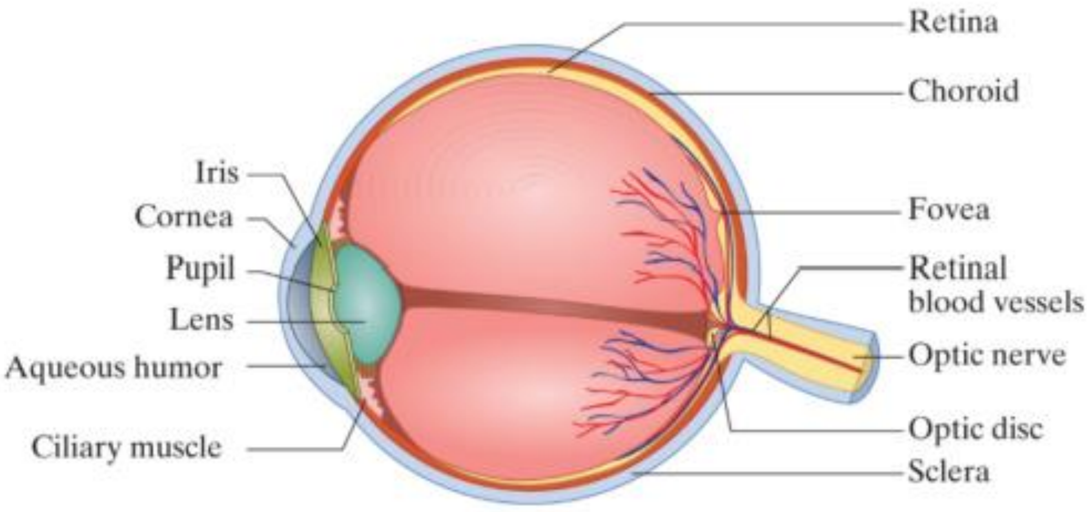
- In the retinal, there are two major Photoreceptor (rods and cones) → responsible for absorbing light (photon)
- Pupil controls: how much light comes in
  - in dark: expands and –in bright light shrinks




Quest: which camera function has the same functionality as the pupil?

# Human Eye: How we see

Figure 2.1 Cross section of the human eyeball.




- In the retinal, there are Photoreceptor (rods and cones)
- Photoreceptor absorbs a light (photon) → photo is captured in retinal → the information is passed through several layers of cells → finally reaches to optic nerve → time to take exit
- The optic nerve **transmits electrical impulses from your eyes to your brain**. Your brain processes this sensory information so that you can see

# Visual Pathway

- Photoreceptor (rods, cones) absorbs a photon → the information is passed through several layers of cells

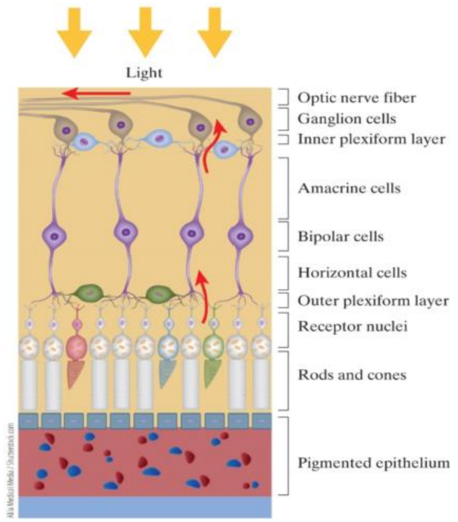



**Figure 2.7** After a photoreceptor absorbs a photon, the information is passed through several layers of cells before exiting via the optic nerve. Surprisingly, the light passes through these same layers of cells before landing on the photoreceptors.

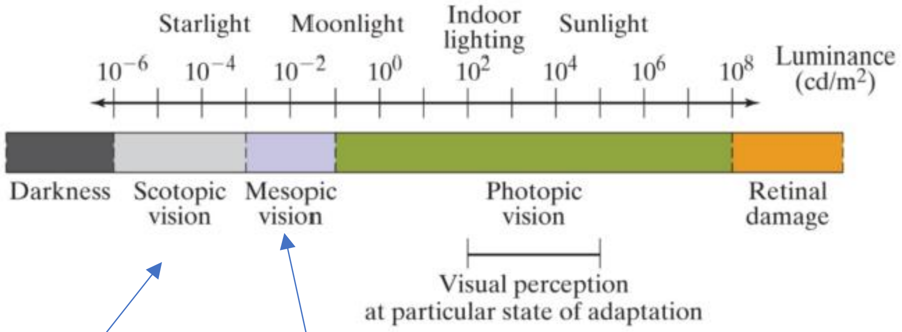
→ finally reaches to optic nerve → time to take exit

- The optic nerve **transmits electrical impulses from eyes to the brain**.
- The brain processes this sensory information so that we can see the image

# Human Visual Perception Under Different Light

Humans can see in this range

Figure 2.8 Scotopic, mesopic, and photopic vision at different light levels. While the human visual system is capable of sensing light in approximately a range of $10^{14}$ overall (from $10^{-6}$ to $10^{8}$ cd/m²), light can be sensed in a range of $10^{3}$, at any particular state of adaptation.




**Question:** Humans can distinguish more variance of gray than more variation of red- why?

Ans: There are more rods than cones [around 10 million: 6 million]

rods are active (used for visualizing black-and-white or gray vision)

cones are active (useful for color visions)

Rods + cones are active in low lights (not in equal ration though)

# Animal Vision

Why to mimic animal vision?

- The imitation of natural systems is known as **biomimicry** (or *biomimetics*).
  - It is an important approach to discovering novel solutions in both software and hardware.
- In a compound eye, the photoreceptors are arranged in small groups called **ommatidia**.
  - Each ommatidium views the world from a different direction, yielding a mosaic of images providing a fairly low-resolution representation of the scene.

Quest: Who uses compound/composite eyes?

# Animal Vision

Figure 2.10 The common housefly has the fastest visual response of any animal, leading to extreme maneuverability in flight. Tiny flying robots (such as this one from Centeye) have been inspired to mimic the housefly's navigation ability based on optic flow.

mrfiza / Shutterstock.com, Courtesy of Centeye

The vision of a fly is more mechanical and less chemical --> chemical reactions take time → therefore fly can easily spot different things faster → the images are low resolution though

# Animal Vision

**Figure 2.11** Raptors, such as this hawk, have the highest visual acuity of any animal. Megapixel video cameras with similar ability are now commercially available.

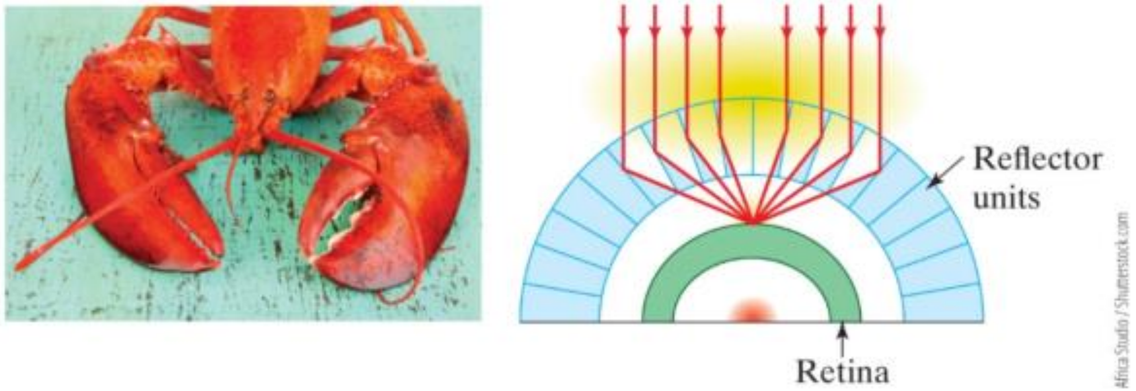

Ronnie Howard / Shutterstock.com, ymgerman / Shutterstock.com

Hawks also have sharp vision → can see spot objects from a significant distance → mimicry makes sense!

# Animal Vision

Figure 2.14 Bees have ultraviolet filters enabling them to detect the flower center, which is helpful for pollinization. The middle image shows the flower (left) as it appears to a bee. Ultraviolet cameras are also used to detect heavenly bodies, such as the sun (right).


Figure 2.16 The lobster eye focuses by reflection, not refraction, and is the inspiration for a new generation of telescope. Based on *Trilobite Eyes—Ultimate Optics* by Kurt Wise https://answersingenesis.org/extinct-animals/trilobite-eyes-ultimate-optics/




How do bees see a flower: bee uses UV filter → enabling them to see the center part → UV cameras are also used to see heavily bodies in space

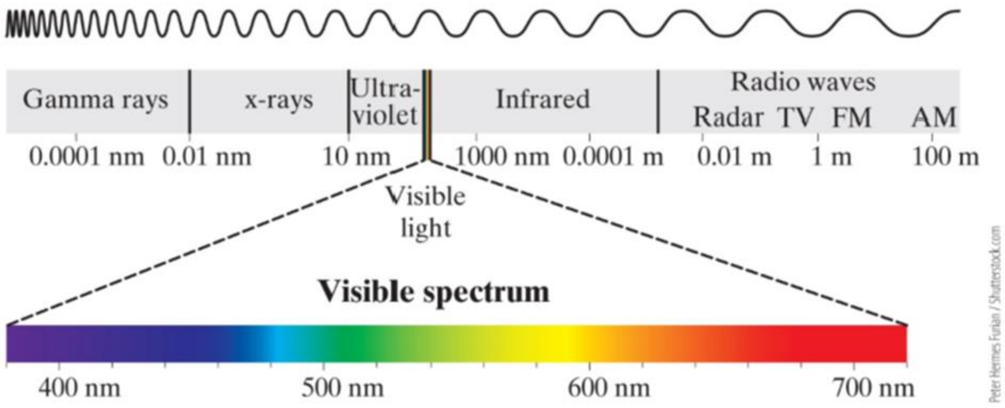
# Light wave

## Image Formation - Light and the Electromagnetic Spectrum

- **Wavelength λ (measured in meters):** the distance between successive peaks in the sinusoid of a wave.
- **Frequency *v* (measured in hertz):** inversely related to the wavelength.
  - **λ** times ***v*** is the speed of light in the medium.

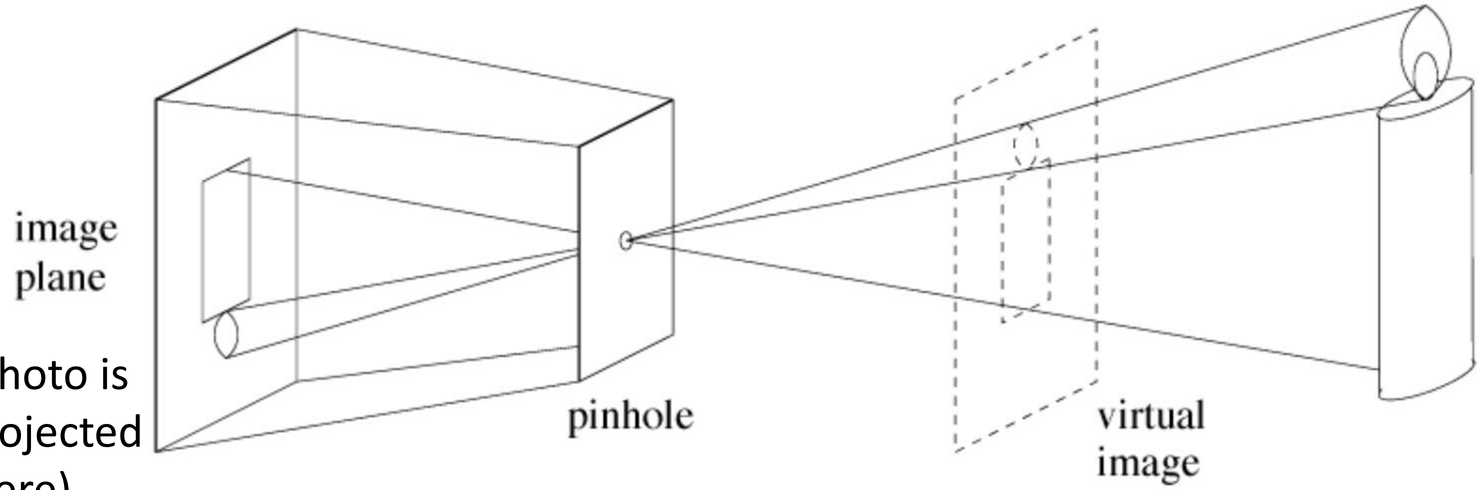
## Electromagnetic Spectrum

**Figure 2.20** The electromagnetic spectrum consists of gamma rays and X-rays at one end, and radio waves and microwaves at the other end. The visible spectrum is between about 380 and 720 nm.



# How We See the Image Through Camera

# What is this?




(photo is
Projected
here)

D. Forsyth, http://luthuli.cs.uiuc.edu/~daf/book/bookpages/slides.html

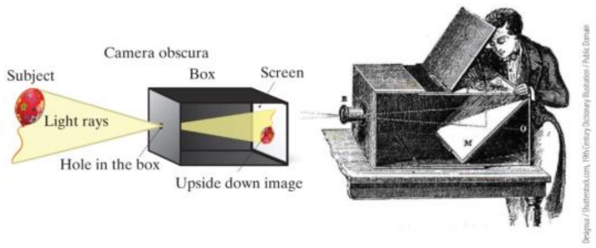
# Pinhole Camera

Quest: you have an image/scene, can you project it on the white wall?

Ans: Yes, by focusing!

- To form a recognizable image, the light rays must be constrained.
  - One way to do this is to construct an empty, opaque box that is so tight that no light can enter the box.
  - Then, a small hole the size of a pin is pierced into one side of the box, which allows light to enter the box only through the hole.

**Figure 2.22** In a pinhole camera, light rays pass through the tiny aperture and form an upside-down image on the opposite wall. A camera obscura was an early form of pinhole camera in which light rays pass through the small aperture, reflect off the mirror, and form an image on the top horizontal surface near the rear of the enclosed box.






**Pinhole camera details:**

- Pinhole is used to create the passing of light in the focused way
- The light is reflected from an object → Can you make sure the light is focused/constrained --> yes, through a pinhole → they come in and projected in the back of the box → result: image!
- Drawback: you need to allow exposure time → does not work instantly

What happens if the pinhole is bigger?

What happens if the pinhole is smaller?

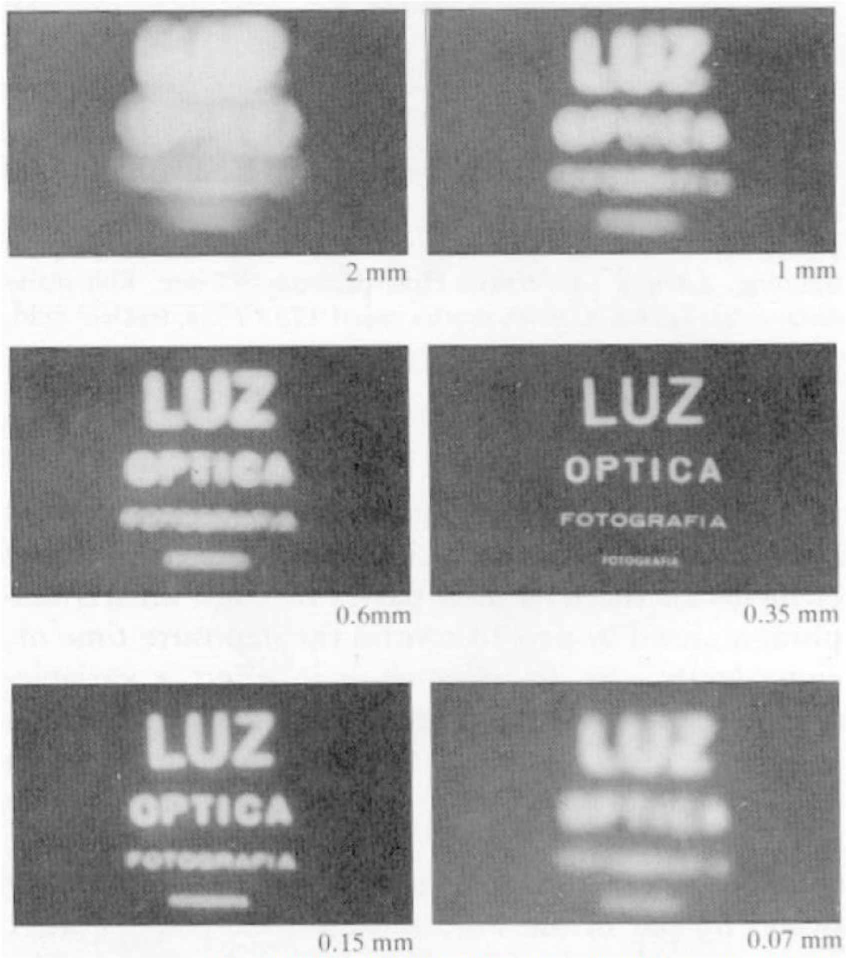
Ans: if bigger, image will be blurry → if smaller, still low quality since enough light not getting in

# Effect of Pinhole Size in Imaging

Pinhole too big - many directions are averaged, blurring the image

Pinhole too small- diffraction effects blur the image

Generally, pinhole cameras are *dark*, because a very small set of rays from a particular point hits the screen.

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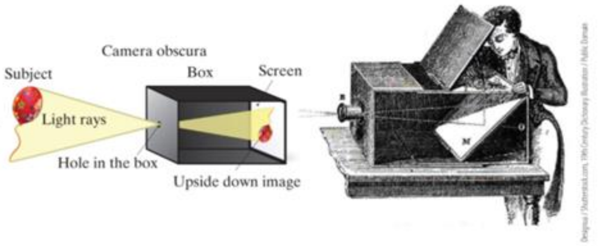




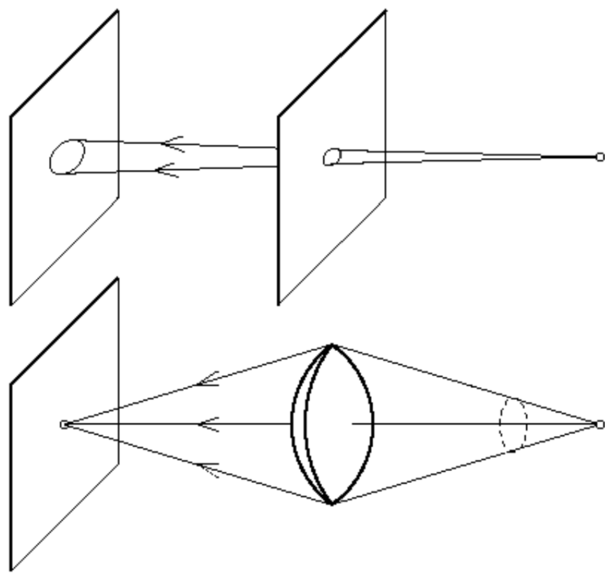

- **Focal point:** the pinhole through which all rays of light pass.
- **Image plane:** the sensor surface on which the image is formed.
- **Optical axis:** the line through the focal point perpendicular to the image plane.
- **Focal length:** the distance from the focal point to the image plane along this line.

# Focusing in Camera

What plays the role of pinhole in cellphone?

- Lenses

## The reason for lenses



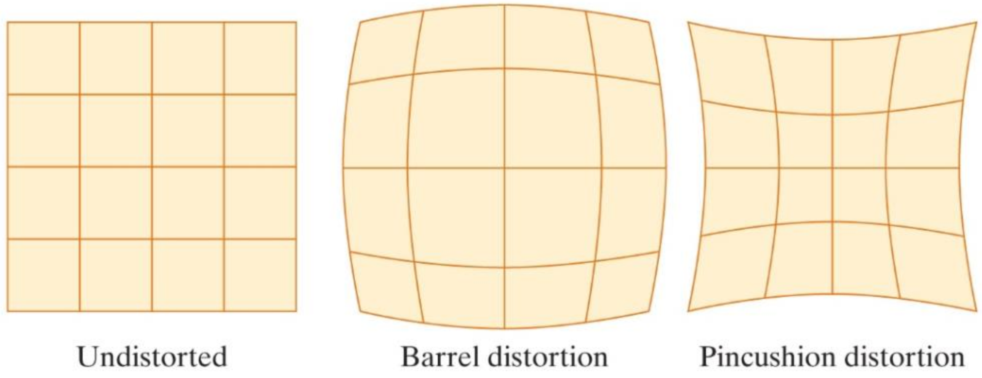
they focus better than pinholes!

# Common Issues with “Camera with Lense”

- **Aberration:** any deviation of the performance of a lens from ideal.

- **Distortion:** arises from the fact that the light rays do not necessarily follow straight lines when passing through the lens.

Figure 2.26 An undistorted image, barrel distortion, and pincushion distortion.

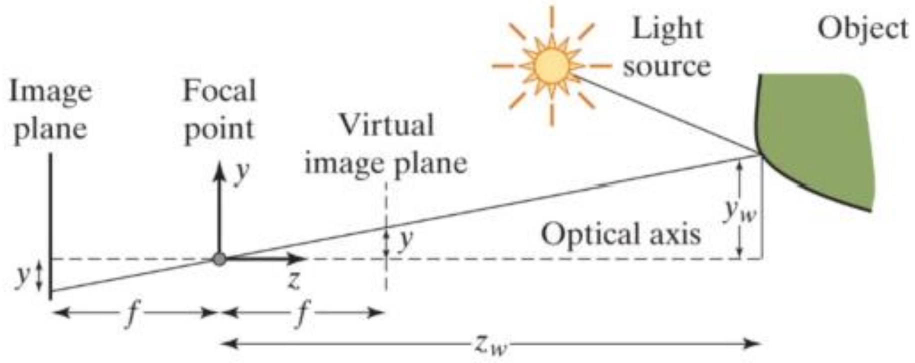


# Basics on How Image in Projected

- **Perspective projection:** Light rays from the source reflect off the surface of an object in the scene, travel through the focal point (also called the center of projection), then land on the image plane.

- $(x_w, y_w, z_w)$: the 3D coordinates of the world point.
- $(x, y)$: the 2D coordinates of its projection onto the image.

$$x = f\frac{x_w}{z_w} \qquad y = f\frac{y_w}{z_w}$$

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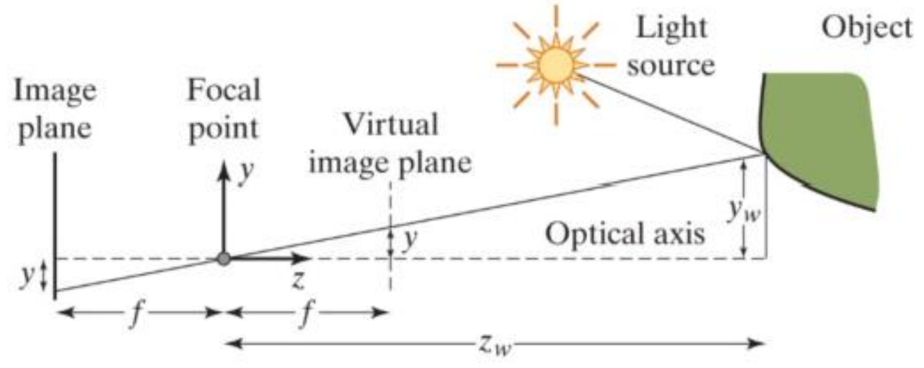
# Perspective projection: worldly coordinates to image coordinates

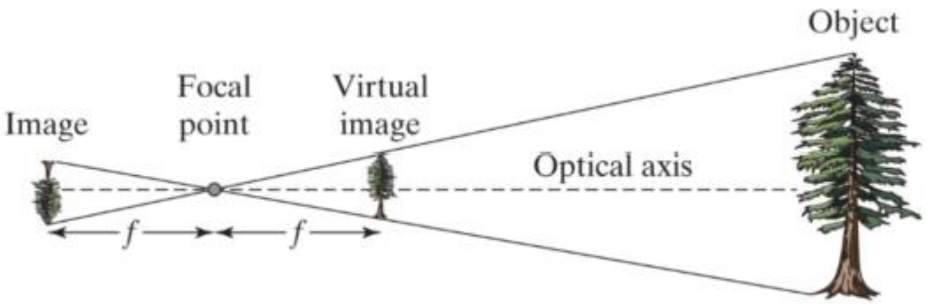
Properties of projection:

- Points go to points
- Lines go to lines
- Planes go to whole image
- Polygons go to polygons
- Degenerate cases
  - line through focal point to point
  - plane through focal point to line

# More correction




What modification is required?



# Image formation overview

Image formation involves

- *geometry* – path traveled by light
- *radiometry* – optical energy flow
- *photometry* – effectiveness of light to produce "brightness" sensation in human visual system
- *colorimetry* – physical specifications of light stimuli that produce given color sensation
- *sensors* – converting photons to digital form

# Basic Imaging Model

Irradiance (light signal = continuous signal) → sampling (discrete signal) → quantization (more discrete)

- The **irradiance** $E$ on the image sensor is the product of a lighting function $\Lambda$ and a reflectance function $R$.

$$E(x, y, \lambda) = \Lambda(x, y, \lambda) R(x, y, \lambda)$$

- The image pixel value $I(x, y)$ can be modeled as the integration of the irradiance function over the area of the pixel and over all wavelengths, after first multiplying by the sensitivity function:

$$I(x, y) = \varphi\left(\int\int\int E(x', y', \lambda') s(\lambda')\, dx'\, dy'\, d\lambda'\right)$$

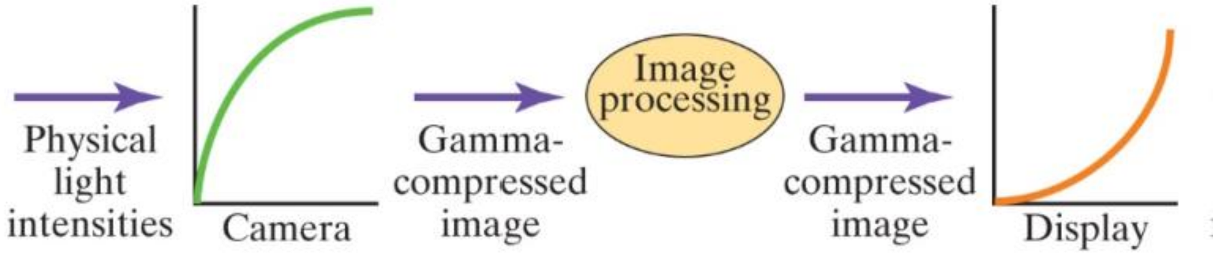
- **Sampling:** converts the continuous irradiance function into a discrete function defined only over the rectangular lattice of integer $(x, y)$ coordinates.

- **Quantization:** assigns a discrete gray level to every pixel in order to represent its value in digital form.

# Tuning the base model: Gamma Compression

- **Gamma Compression:** transforms the linear, physical light intensity into a perceptually uniform quantity, so that the pixel values in a digital image are not directly proportional to the amount of light collected by the sensor.

Figure 2.29 Linear light intensities are gamma compressed by the camera into perceptually uniform quantities, which are then gamma expanded by the display.



# Tuning the base model: Contrast adjustment

Which inner square is brighter?

**Figure 2.31** Simultaneous contrast. The pixels inside the middle squares have the same luminance, but the pixels on the right appear brighter due to its surroundings. Therefore, if an image is displayed at the correct luminance in a dimmer environment than the one in which it was captured, it will appear to be lacking in contrast.

# Sensors: (self study)

## CCD and CMOS Sensors

- **CCD and CMOS Sensors:** each consists of a dense array of photodiodes (typically spaced 1 to 5 μm apart) that convert light photons to electrons.
- **CCD sensor:** electrons are collected and stored in local potential wells during exposure time and then read out by transferring electrons down the line of potential wells until they reach the readout register known as the *horizontal shift register*.
- **CMOS sensor:** transistors next to each photodiode convert the electrons to a voltage.

# Storage of Videos: (self study)

## Transmission and Storage

- Traditional video cameras transmit the video signal using analog waveforms. The three standards are:
  - NTSC: used in North America and Japan
  - SECAM: used in France and the former Soviet Union
  - PAL: used most everywhere else
- **Composite video:** the luminance and chrominance information is combined into a single signal.
- **Component video:** individual color channels are separated into individual signals, resulting in higher fidelity.

# Medical Imaging (self study)

- **X-ray radiography:** produces images by transmission rather than reflection.
- A generator emits X-rays toward an object of interest, and a detector measures the photons that make it to the other side, rather than being absorbed by the object.

- **Magnetic resonance imaging (MRI):** safer than X-ray because it does not use ionizing radiation.
  - MRI uses powerful magnets to align the hydrogen nuclei (that is, protons) in the water molecules.

- **Positron emission tomography (PET):** another technique for nuclear imaging.
  - A patient is injected with a radioactive isotope which, as it undergoes positron emission decay, emits a positron.

# Remote Sensing (self study)

- Cameras are often attached to aircraft or satellites for remote sensing of the earth for applications in meteorology, agriculture, surveillance, and geology.
  - These cameras typically sense multiple frequencies simultaneously.